SKIN HEALTH KNOWLEDGE BASE

Evidence-Informed Guide to Acne, Comedones, Cysts, Dark Spots, Enlarged Pores and Wrinkles

Medical treatment overview • Daily care • Skincare routines • Nutrition plan • Safety and referral criteria

Document field	Details
Purpose	Structured educational content for retrieval-augmented generation (RAG), chatbot answers, and patient education.
Audience	General adult audience; not a substitute for individual medical assessment.
Topics covered	Acne; blackheads and whiteheads; cysts/nodules; dark spots; papules and pustules; visible pores; wrinkles.
Evidence approach	Guidance from the American Academy of Dermatology (AAD), NHS, and published clinical guidance.
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Important medical notice

This document is general educational information. It cannot diagnose a skin lesion or determine which medicine is safe for a specific person. Skin conditions that look similar can have different causes. A clinician or dermatologist should assess persistent, painful, scarring, rapidly changing, infected-looking, or uncertain lesions.

Do not start prescription medicines, oral antibiotics, isotretinoin, hormonal medicines, steroid creams, skin-lightening medicines, or high-dose supplements based only on this guide. Pregnancy, breastfeeding, age, allergies, medical history, and current medicines can change treatment choices.

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How to use this guide

Read the safety principles first. The condition-specific chapters describe common options and limitations. The routines and meal framework are examples, not mandatory prescriptions. In a RAG system, retrieve the relevant condition section plus safety caveats rather than presenting every treatment to every user.

1. Executive summary and safety principles

1.1 Core principles

- Correctly identify the concern before suggesting treatment. A blackhead, an inflamed acne papule, an epidermoid cyst, rosacea, folliculitis and other lesions may be confused in casual descriptions.
- Keep skincare simple: gentle cleansing, moisturizer, sunscreen, and one targeted active ingredient introduced gradually are usually more sustainable than many simultaneous products.
- Acne treatment commonly requires consistent use for several weeks; meaningful improvement often takes 6–8 weeks, and a fuller assessment is often made around 12 weeks.
- Do not squeeze, pick, lance, or attempt to drain acne lesions or cysts. This can worsen inflammation, infection risk, pigment changes and scarring.
- Daily broad-spectrum SPF 30+ sunscreen is important for preventing sun damage and helping dark marks from becoming darker.
- Food can influence acne in some people, but no diet reliably cures acne. Prefer balanced, minimally processed meals; avoid unnecessary restrictive diets.
- Deep, painful, scarring or widespread acne and suspected cysts warrant clinician review rather than relying on home care.

1.2 What is and is not a skincare goal

Concern	Realistic goal	Important limitation
Acne / papules / pustules	Reduce new lesions, inflammation and scarring risk.	A treatment may cause early dryness; improvement is gradual.
Blackheads / whiteheads	Reduce clogged pores and prevent new comedones.	Pore strips and squeezing offer temporary results and may irritate.
Cyst-like lump	Confirm diagnosis; reduce pain and prevent scarring/infection.	A true cyst may need medical treatment or a procedure.
Dark spots	Prevent new marks and gradually fade existing pigmentation.	Some marks take months; irritation can make pigmentation worse.
Visible pores	Make pores appear less noticeable by controlling oil and congestion.	Pores are normal structures and cannot be permanently erased.
Wrinkles	Prevent additional UV damage and soften the appearance of fine lines.	Creams cannot reproduce surgical results or completely remove deep folds.

1.3 Safety boundaries

Seek urgent medical advice for rapidly spreading redness, warmth, swelling, severe or escalating pain, fever, significant pus with systemic illness, eye-area swelling, or a rapidly changing or bleeding lesion. Seek prompt assessment for deep painful acne, scarring, recurring lumps, or a new lesion that does not behave like ordinary acne.

2. Skin anatomy and terminology

2.1 Why pores become blocked

A pore is the opening of a hair follicle and its associated oil gland. Acne develops when dead skin cells and sebum accumulate in the follicle and inflammation occurs. Hormones, genetics, medications and other factors can contribute. Acne is not simply caused by poor hygiene.

2.2 Common lesion terms

Term	Plain-language description	Typical interpretation
Comedone	A clogged follicle.	Can be open (blackhead) or closed (whitehead).
Blackhead	An open comedone with a dark surface.	The colour is not dirt; material in the pore darkens on exposure to air.
Whitehead	A closed or skin-coloured/white comedone.	A blocked follicle without an open surface.
Papule	Small raised inflamed bump, usually without visible pus.	May be tender or red/brown depending on skin tone.
Pustule	Inflamed bump with visible pus.	Do not squeeze; it may leave a mark or scar.
Nodule	Deeper, firm, often painful inflammatory lesion.	Higher risk of scarring; clinician assessment is advised.
Cyst-like acne lesion	Deep, tender lump that may feel fluid-filled.	Terminology is often used loosely; diagnosis needs examination.
Post-inflammatory hyperpigmentation (PIH)	Flat dark mark left after inflammation or injury.	Common after acne, particularly in medium-to-deep skin tones.
Atrophic/hypertrophic scar	Indented or raised permanent change in skin texture.	Different from a flat dark mark; treatment options differ.

2.3 Skin types and tolerance

- Oily skin can still be dehydrated or irritated; do not over-cleanse to remove every trace of oil.
- Dry or sensitive skin may need fragrance-free moisturizer and a slower introduction of active ingredients.
- Darker skin tones can be more prone to visible post-inflammatory dark marks. Irritation from strong acids, scrubs or unregulated bleaching products may worsen them.
- A product labelled non-comedogenic is less likely to clog pores, but no label guarantees that it will suit every person.

2.4 Patch testing

For a new non-prescription skincare product, test a small area according to the product directions for several days before wider use. Stop using it if persistent burning, swelling, rash or marked irritation occurs. Patch testing does not guarantee that a product will never cause a reaction.

3. Acne vulgaris: causes, severity and treatment pathways

3.1 Contributing factors

- Excess sebum and follicular blockage.
- Inflammation and changes in the follicular microbiome.
- Hormonal influences, including puberty and some adult hormonal patterns.
- Genetic predisposition; acne may run in families.
- Some medicines and products can worsen acne in certain people.
- Friction, occlusive equipment, hair products, picking and irritation may contribute in some cases.

3.2 General treatment framework

Presentation	General approach	Escalation
Occasional mild comedones or pimples	Gentle routine plus one OTC active such as benzoyl peroxide or salicylic acid, used as directed.	Ask a pharmacist/clinician if uncertain or irritated.
Persistent mixed comedonal and inflammatory acne	A clinician may recommend combination treatment, often a topical retinoid with another agent.	Review adherence and response after a consistent trial.
Moderate-to-severe inflammatory acne	Clinician-directed topical combinations and sometimes oral treatment.	Oral antibiotics should be limited in duration and generally paired with topical therapy, including benzoyl peroxide where appropriate.
Deep nodules, cyst-like lesions, scarring or major distress	Dermatology assessment; prescription options may be needed.	Severe or treatment-resistant acne may warrant isotretinoin under specialist monitoring.

3.3 Common topical ingredients

Ingredient	Role	Cautions
Benzoyl peroxide	Reduces acne-associated bacteria and inflammatory lesions.	Can dry/irritate skin and bleach fabrics; start with a tolerable formulation.
Adapalene / other topical retinoids	Help unclog follicles and prevent comedones; some also improve acne-related dark marks.	Irritation and peeling can occur. Avoid topical retinoids during pregnancy or when trying to conceive unless a treating clinician gives specific advice.
Salicylic acid	Helps exfoliate and unclog pores; may help blackheads and whiteheads.	Can irritate, especially when layered with other actives.
Azelaic acid	Can help acne inflammation and post-acne pigmentation.	May sting or irritate; concentration and suitability matter.
Topical antibiotics	May be prescribed for inflammatory acne.	Do not use antibiotic monotherapy; resistance prevention is important. Use only under clinician guidance.

3.4 Prescription-only or clinician-supervised options

Depending on severity, medical history and local availability, a clinician may consider oral doxycycline or another antibiotic, hormonal therapy for eligible patients, or oral isotretinoin for severe, scarring or treatment-resistant acne. These are not self-treatment options. Oral isotretinoin has major pregnancy-prevention requirements and needs clinical monitoring. Antibiotics can have important adverse effects and interactions.

3.5 Practical use principles

- Use acne treatments consistently as directed, often on the acne-prone area rather than only dabbing visible spots, if the product instructions say so.
- Introduce one active ingredient at a time. Avoid starting a retinoid, strong acid, benzoyl peroxide and exfoliating scrub together.
- Use moisturizer and sunscreen to support tolerance.
- If irritation is significant, stop the irritating product and seek professional advice; do not push through severe burning or swelling.

4. Blackheads and whiteheads (comedonal acne)

4.1 How they differ

Blackheads are open comedones; the material at the opening darkens and is not simply dirt. Whiteheads are closed comedones. Both are forms of non-inflammatory acne, though surrounding skin can become irritated.

4.2 Care options

- A topical retinoid such as adapalene is commonly used to prevent clogged pores; ask a clinician or pharmacist whether it is suitable for you.
- Salicylic acid may help loosen and clear pore congestion. Start according to the label and avoid excessive layering.
- Use a gentle cleanser and a non-comedogenic moisturizer; oil-free or non-comedogenic sunscreen may be useful for acne-prone skin.
- Avoid harsh scrubs, repeated pore strips, metal extractors at home and squeezing. These can injure skin and worsen marks.

4.3 Example gradual routine (general adult, non-pregnant)

Time	Example	Notes
Morning	Gentle cleanser → moisturizer → broad-spectrum SPF 30+.	If using benzoyl peroxide, follow the product or clinician instructions; it may bleach fabric.
Evening	Cleanser → one selected treatment (e.g., retinoid or salicylic acid) → moisturizer.	Start slowly according to label/clinician advice; do not layer multiple new actives.
First 2 weeks	Monitor dryness, redness and stinging.	Reduce frequency or pause if irritation develops; ask a professional if severe.
Weeks 6–12	Assess whether new comedones are decreasing.	Seek review if no meaningful improvement or if painful lesions/scars develop.

4.4 What to avoid

- Toothpaste, lemon juice, baking soda, undiluted essential oils or household chemicals on skin.
- Aggressive exfoliation or scrubbing until the skin feels squeaky clean.
- Picking at blackheads or trying to extract deep lesions.
- Changing the entire routine every few days before a fair trial.

5. Cysts and deep painful acne lesions

5.1 Important distinction

People often call any large bump a 'cyst', but a deep acne nodule, an epidermoid cyst, an abscess, an inflamed hair follicle and other conditions can look similar. A true cyst may have a central opening or recur in the same place, but appearance alone is not enough to diagnose it.

5.2 Why assessment matters

- Deep acne nodules and cyst-like lesions have a higher risk of permanent scarring and pigment changes.
- A skin abscess or infected cyst may require medical drainage or other treatment; squeezing it at home can worsen infection.
- A dermatologist may use prescription acne therapy, and in selected cases may inject a corticosteroid into a large inflamed acne lesion to reduce inflammation.

5.3 Safe interim care

- Do not squeeze, puncture, cut, needle or attempt to drain the lump.
- Use a gentle cleanser and avoid acids or retinoids directly over broken or severely inflamed skin unless a clinician directs otherwise.
- A clean warm compress may provide comfort for some superficial inflamed lumps, but it does not replace assessment and should not be hot enough to burn.
- Arrange medical review for a painful, enlarging, recurrent or persistent lump, especially on the face.
- Seek urgent care if redness spreads, the area becomes hot and swollen, pain rapidly worsens, fever develops, or you feel unwell.

5.4 Questions a clinician may ask

- When did it begin, and is it growing or recurring in the same location?
- Is it painful, warm, draining, or associated with fever?
- Are there multiple similar lesions or acne elsewhere?
- What skincare products and medicines are being used?
- Has the lesion been squeezed, punctured or treated before?

5.5 Do not self-prescribe

Do not use leftover antibiotics, oral isotretinoin, steroid tablets, steroid creams or prescription-strength combinations without a clinician. The correct treatment depends on whether the lesion is acne, a cyst, an abscess or another condition.

6. Papules and pustules

6.1 Recognition

Papules are inflamed raised bumps without obvious pus; pustules have a visible pus-filled centre. They may occur alongside blackheads and whiteheads. Redness may appear brown, purple or darker than surrounding skin rather than bright red in deeper skin tones.

6.2 General care

- A gentle cleanser, non-comedogenic moisturizer and sunscreen form the foundation.
- Benzoyl peroxide can help inflammatory acne; lower-strength products may be easier to tolerate. Follow label directions.
- A topical retinoid may be used to treat comedones and prevent new lesions, often as part of a combination plan.
- Azelaic acid may be useful when acne and post-inflammatory dark marks coexist, subject to tolerance and professional advice.
- Avoid topical antibiotic monotherapy. Prescription antibiotics should be selected and monitored by a clinician.

6.3 Reduce marks and scars

- Do not pick or pop pustules.
- Treat acne early enough to reduce ongoing inflammation.
- Use daily sunscreen; UV and visible light can deepen some dark marks.
- Ask for dermatology advice if lesions are painful, widespread, recurring, leaving scars or affecting confidence and mood.

6.4 Irritation versus allergy

Mild dryness or brief stinging may occur with some actives, especially early on. Persistent burning, marked swelling, blistering, hives or breathing difficulty are not expected routine effects; stop the product and seek appropriate medical help. Breathing difficulty or facial/throat swelling is an emergency.

7. Dark spots and post-inflammatory hyperpigmentation

7.1 Common causes

Flat brown, grey-brown or darker marks after acne, scratching, eczema, injury or irritation may be post-inflammatory hyperpigmentation (PIH). Other causes of pigmentation exist, including melasma and some medical conditions, so persistent or unusual patches may need assessment.

7.2 First step: stop the trigger

- Control the acne or inflammation causing new marks.
- Avoid picking and prevent repeated irritation from scrubs, strong acids or unregulated skin-lightening products.
- Use broad-spectrum SPF 30+ every day. For persistent pigmentation, a tinted sunscreen containing iron oxides may add protection against visible light, if suitable for the user.

7.3 Ingredients a clinician may consider

Option	Potential role	Cautions
Azelaic acid	May improve acne and some dark marks.	Can sting or irritate; introduce carefully.
Topical retinoid	May improve acne and help some pigmentation over time.	Irritation can worsen PIH; avoid during pregnancy/trying to conceive unless clinician specifically advises.
Hydroquinone	Can be effective for selected pigment disorders.	Use only with appropriate clinician guidance; unsuitable use can cause harm or further discoloration.
Other prescription or procedural options	May be selected for specific diagnoses.	Peels, lasers and procedures can worsen pigmentation if poorly matched to skin type or performed inappropriately.

7.4 Expectations

Some superficial marks fade gradually over months; deeper pigmentation may take much longer. The best plan prevents new marks while allowing existing ones to fade. Rapid bleaching is not a safe or realistic target.

7.5 Special caution

Avoid products with unknown ingredients, mercury-containing creams, unsupervised potent steroids, and unlabelled 'whitening' mixtures. Do not apply lemon juice or household bleaching products. Seek assessment if pigmentation is sudden, spreading, associated with other symptoms, changing in shape/colour, or not clearly linked to a prior spot.

8. Enlarged pores and skin texture

8.1 What pores are

Pores are normal openings in the skin. Their visibility is influenced by genetics, oil production, comedones, skin elasticity and sun damage. A routine can reduce congestion and sometimes make pores appear less prominent, but pores cannot be permanently closed or erased.

8.2 Practical routine

- Cleanse gently, generally no more than twice daily unless sweating or product directions require otherwise.
- Consider salicylic acid for oily, congested skin if tolerated.
- A topical retinoid may help comedones and skin texture over time, when appropriate.
- Moisturize even if skin is oily; dehydration and irritation can make texture more noticeable.
- Use SPF 30+ daily to limit UV-related collagen damage.

8.3 Avoid misleading claims

- Cold water, ice or a toner cannot permanently shrink pores.
- Pore strips may remove surface material temporarily but do not prevent pores from refilling.
- Harsh cleansing and frequent exfoliation may worsen oiliness perception and irritation.
- Professional procedures may help selected texture concerns, but need an individualized assessment, especially for darker skin tones due to pigment-change risk.

9. Wrinkles and photoaging

9.1 Causes and realistic goals

Fine lines and wrinkles reflect intrinsic aging, repeated facial movement, reduced skin elasticity and cumulative ultraviolet exposure. Smoking and tanning can accelerate visible aging. Skincare can help prevent additional damage and soften some fine lines, but cannot fully remove deep folds.

9.2 Evidence-informed daily care

- Apply broad-spectrum SPF 30+ sunscreen to exposed skin daily; reapply according to label directions, especially outdoors, after sweating or swimming.
- Use a moisturizer. By trapping water, it can temporarily make fine lines look less prominent.
- Avoid tanning beds and deliberate tanning; use protective clothing and shade.
- Avoid smoking and maintain a balanced diet and adequate sleep as part of general health.

9.3 Retinoids and anti-aging products

Topical retinoids may improve fine lines and texture for some people, but can cause dryness, redness and peeling. Start only if appropriate, follow product or clinician instructions, and introduce gradually. Do not use topical retinoids during pregnancy or while trying to conceive unless a treating clinician gives specific advice. OTC cosmetic retinol products vary in strength and evidence.

9.4 Procedures

Prescription treatments and procedures such as chemical peels, microneedling, lasers, injectables or surgery have different indications, risks, costs and recovery times. They should be discussed with a qualified clinician. Inexperienced procedures can cause burns, scars, infection or persistent pigment changes.

9.5 Product-selection rules

- Choose fragrance-free products if sensitive; 'natural' does not automatically mean safer.
- Introduce one new anti-aging product at a time.
- Stop if a product causes persistent irritation; irritation does not mean it is working better.
- Be sceptical of claims promising permanent pore closure, instant wrinkle removal or guaranteed results.

10. Daily skincare routines and product selection

10.1 Basic morning routine

1. Cleanser: gentle, non-abrasive, suitable for skin type. Avoid scrubbing.
2. Optional treatment: only a selected ingredient appropriate to the concern and tolerated by the skin.
3. Moisturizer: preferably fragrance-free and non-comedogenic if acne-prone.
4. Sunscreen: broad-spectrum SPF 30 or higher; choose a comfortable formula so it can be used consistently.

10.2 Basic evening routine

5. Remove makeup and sunscreen gently; do not sleep in makeup.
6. Cleanse with lukewarm water and a gentle cleanser.
7. Apply the chosen acne/pigmentation treatment according to label or clinician instructions.
8. Apply moisturizer. If using a retinoid and irritation is an issue, ask a clinician whether moisturizer-first or less frequent application is appropriate.

10.3 Example weekly schedule for introducing one active

This is a conservative illustration, not a prescription. The exact frequency depends on product strength, local instructions, age, skin sensitivity, pregnancy status and clinician advice.

Period	Approach	Check
Days 1–3	Basic cleanser + moisturizer + sunscreen; patch test new product.	Any rash, swelling or marked sensitivity?
Week 1	Use one selected active only as directed; do not add other new actives.	Mild dryness versus persistent burning.
Weeks 2–3	Continue if tolerated; use moisturizer and sunscreen consistently.	Reduce or stop and seek advice if irritation is significant.
Weeks 4–8	Keep routine stable; note new lesions and tolerance.	Avoid judging success by one day or one breakout.
Weeks 8–12	Review response and adherence; seek professional review if inadequate.	Painful lesions, scarring or worsening means earlier review.

10.4 Product labels and ingredients

- Prefer non-comedogenic products if acne-prone; fragrance-free products may suit sensitive skin.
- Avoid stacking multiple exfoliating acids, retinoids and benzoyl peroxide without a clear plan.
- Check expiry dates, packaging and ingredient lists; buy from reputable sources.
- Do not share prescription creams or use someone else's medication.

10.5 Common mistakes

- Washing the face repeatedly to remove oil.
- Changing products every few days.
- Using scrubs, brushes or home extraction tools on inflamed acne.
- Skipping moisturizer because the skin is oily.
- Using sunscreen only at the beach rather than during routine daytime exposure.
- Expecting diet or skincare alone to resolve deep cystic or scarring acne.

11. Nutrition and practical 7-day meal framework

11.1 What evidence can support

A balanced diet supports general health, but there is no single 'acne diet'. Some small studies suggest a lower-glycemic eating pattern may help acne in some people. Evidence linking dairy to acne is mixed; do not remove entire food groups without a reason or nutritional replacement. Food choices should be adapted to allergies, diabetes, kidney disease, pregnancy, cultural preferences, budget and clinician advice.

11.2 Practical principles

- Build meals around vegetables, whole fruit, legumes, whole grains and adequate protein.
- Choose water or unsweetened drinks more often than sugary beverages.
- Reduce frequent large portions of highly refined carbohydrates and sweets if they dominate the diet; replace rather than simply restrict.
- Include protein sources such as eggs, fish, chicken, beans, lentils, tofu or other suitable options.
- Include healthy fats from nuts, seeds, olive/canola oils or fish, according to preference and availability.
- If a specific food seems to trigger acne, keep a simple food/skin diary and discuss a balanced trial with a clinician rather than making broad exclusions.
- Do not take high-dose vitamin A, zinc or other supplements for skin without advice; excess can be harmful and supplements are not a replacement for acne treatment.

11.3 Sample South Asian-style 7-day framework

Portions should match appetite, activity, age and health needs. This is a flexible example, not a calorie-controlled weight-loss plan.

Day	Breakfast	Lunch	Dinner	Snack option
1	Egg or chickpea omelette + whole-wheat roti	Daal, sabzi, salad + moderate roti/rice portion	Grilled fish/chicken or beans + vegetables	Whole fruit or unsalted nuts
2	Plain oats with nuts/seeds and fruit	Chicken/bean bowl with brown or mixed-grain rice and salad	Daal + mixed vegetables + roti	Carrots/cucumber with hummus or chana
3	Plain yogurt if tolerated + oats/fruit, or fortified alternative	Chana/rajma + salad + roti	Egg, tofu or chicken with vegetables	Fruit
4	Besan chilla with vegetables	Fish/chicken/beans + vegetables + whole grain	Vegetable daal + roti	Small handful of nuts
5	Whole-grain toast/roti + egg	Leftover daal/beans + salad	Home-cooked chicken or tofu + vegetables + rice	Roasted chana
6	Oats or whole-grain cereal with unsweetened milk/alternative	Mixed lentils + sabzi + roti	Fish/egg/beans + salad and whole grain	Fruit
7	Vegetable omelette or chilla	Home-cooked protein + vegetables + roti/rice	Light daal/soup + vegetables + protein	Plain yogurt if tolerated or fruit

11.4 Foods to limit, not automatically ban

- Sugary drinks, frequent sweets, pastries and heavily refined snack foods, especially if they displace balanced meals.

- Frequent deep-fried foods as part of an overall high-energy, low-fibre eating pattern; greasy food itself is not proven to be a universal direct acne cause.
- If acne appears linked to cow's milk or whey supplements, discuss a time-limited, nutritionally adequate trial with a clinician/dietitian. Evidence is not conclusive and dairy affects people differently.

11.5 Hydration and lifestyle

Drink regularly according to thirst, weather and activity; there is no fixed water amount that clears acne. Sleep, exercise and stress management support wellbeing but should not be presented as guaranteed acne cures. After sweating, rinse or shower when practical and avoid tight, irritating equipment remaining against the skin.

12. Treatment timelines, monitoring and troubleshooting

12.1 What to track weekly

- Number of new inflamed spots and comedones (estimate, not perfection).
- Pain, tenderness, drainage or recurring deep lumps.
- New dark marks or scars.
- Dryness, burning, peeling, redness or rash.
- Products used, frequency, and missed applications.
- Possible triggers, menstrual/hormonal patterns if relevant, and any new medicines.

12.2 Expected timelines

Concern / intervention	Typical expectation	When to review
OTC acne treatment	May need 6–8 weeks to see improvement; full assessment often around 12 weeks.	Earlier if severe irritation, worsening, nodules or scarring.
Dark spots	Often fade over months; deeper marks can take longer.	Review if diagnosis is uncertain or marks are changing.
Moisturizer	May improve dryness and temporary appearance of fine lines relatively quickly.	Switch/seek advice if rash or persistent irritation.
Wrinkle-prevention routine	Prevention is cumulative; cosmetic changes are gradual and modest.	Discuss prescription products or procedures with a clinician.
Cyst-like lump	Cannot reliably predict without diagnosis.	Prompt assessment if painful, enlarging, recurrent, hot, draining or persistent.

12.3 If treatment is not working

9. Check whether the product is appropriate for the concern and is being used according to instructions.
10. Check consistency and whether irritation has forced frequent stopping.
11. Avoid adding several new products at once; simplify the routine if irritated.
12. Ask a pharmacist or clinician to review products and medicines.
13. Seek dermatology assessment for deep painful lesions, scarring, widespread acne, uncertain cysts or inadequate response after an appropriate trial.

12.4 Mental wellbeing

Acne and visible skin changes can affect confidence, social life and mood. This impact is valid and is a reason to seek medical help, even if the number of lesions appears small. If skin concerns cause severe distress, social withdrawal or thoughts of self-harm, seek appropriate mental-health support promptly.

13. Red flags and when to seek medical care

Situation	Recommended action
Rapidly spreading redness, heat, swelling, severe pain, fever or feeling systemically unwell	Urgent medical assessment; possible infection needs examination.
Swelling near the eye, vision symptoms, or severe facial swelling	Urgent assessment.
Deep painful nodules/cysts, new scars, or extensive face/chest/back acne	Arrange prompt clinician/dermatologist review.
A lump that recurs in the same place, grows, drains or does not resolve	Medical assessment to identify cyst, abscess or another cause.
A lesion that changes rapidly, bleeds without clear cause, ulcerates or does not heal	Prompt clinical examination; do not assume it is acne.
Persistent acne despite a consistent appropriate treatment trial	Review diagnosis, routine and prescription options with a clinician.
Pregnant, trying to conceive or breastfeeding and considering retinoids/acne medicines	Consult a clinician before using active medicines; avoid self-prescribing.
Severe rash, blisters, facial/throat swelling or breathing difficulty after a product	Stop the product and seek urgent/emergency help as appropriate.

13.1 What to prepare for an appointment

- List of all skincare products, medicines and supplements with frequency.
- Approximate start date and progression of the problem.
- Photos taken in consistent lighting over time, if comfortable.
- History of allergies, pregnancy/trying-to-conceive status where relevant, and prior treatments.
- Main concerns: pain, scars, dark marks, cost, sensitivity or impact on quality of life.

13.2 What not to do before assessment

- Do not puncture, squeeze or drain a lump.
- Do not apply steroid creams, leftover antibiotics, oral acne medication or unlabelled whitening mixtures without medical advice.
- Do not delay care for rapidly worsening symptoms while trying home remedies.

14. RAG-ready knowledge design and sample answer rules

14.1 Recommended metadata fields

Field	Purpose	Example
document_title	Identify source document.	Skin Health Knowledge Base
topic	Primary retrieval category.	comedonal_acne
subtopic	Specific intent.	blackheads_whiteheads
content_type	Separate advice types.	routine / treatment / diet / safety
audience	Scope.	general_adult_education
risk_level	Route high-risk questions safely.	high for cysts, isotretinoin, infection red flags
evidence_source	Traceability.	AAD / NHS / clinical guideline
last_reviewed	Maintenance.	2026-10-28
contraindication_tags	Safety filters.	pregnancy_retinoid_caution
keywords	Improve retrieval.	comedone, blackhead, whitehead, clogged pore

14.2 Suggested chunking strategy

- Chunk by meaningful section, not arbitrary page length. Keep a condition's definition, first-line care and safety caveat together where possible.
- Use approximately 300–700 words per chunk as a starting point, with modest overlap where a heading or safety instruction would otherwise be separated. Tune using retrieval evaluation.
- Preserve headings and lists; store source URLs and review date as metadata.
- Do not mix different conditions into a single treatment chunk. A cyst chunk should not be retrieved as though it were routine comedonal acne.
- Keep contraindications and red flags retrievable alongside any treatment suggestion.

14.3 Example chunk labels

chunk_id	topic	intent	risk
SKIN-001	general_safety	scope_and_disclaimer	high
SKIN-ACNE-010	acne	OTC_treatment_overview	medium
SKIN-COMEDONE-020	blackheads_whiteheads	routine	low-medium
SKIN-CYST-030	cysts_nodules	triage_and_do_not_drain	high
SKIN-PIH-040	dark_spots	sun_protection_and_options	medium
SKIN-DIET-050	nutrition	balanced_meal_framework	low
SKIN-RED-FLAG-060	urgent_care	infection_or_changing_lesion	high

14.4 Guardrails for a skincare RAG assistant

- Start by stating that the response is educational and not a diagnosis.
- Ask a short clarifying question when age, pregnancy status, lesion location, severity or duration materially changes safety.
- Do not diagnose a cyst from text alone; recommend examination for deep, painful, recurrent or enlarging lumps.
- Do not prescribe dosing schedules for oral antibiotics, isotretinoin, hormonal medicines or prescription-only treatments.
- Do not advise topical retinoids in pregnancy/trying to conceive; direct the user to a clinician for suitable alternatives.
- Do not recommend squeezing, puncturing or home drainage.
- Use diet language cautiously: possible associations are not proof that a food causes acne in every person.

- Give red flags early when symptoms suggest infection, rapid change, scarring or severe distress.
- Prefer generic ingredient names and explain that availability, concentration and local regulations vary.
- Include a source citation or source URL in answers when the system supports citations.

14.5 Sample safe RAG answer: blackheads

"Blackheads are open clogged pores, not simply dirt. A gentle cleanser, non-comedogenic moisturizer and daily SPF 30+ are a good foundation. Salicylic acid or a topical retinoid may help, but introduce one product at a time and follow its instructions because irritation can worsen dark marks. Avoid squeezing or aggressive scrubbing. If lesions are painful, scarring, or not improving after a consistent treatment trial, speak with a clinician."

14.6 Sample safe RAG answer: painful cyst-like bump

"A deep painful bump may be a nodule, cyst, abscess or another condition, and text alone cannot confirm which. Do not squeeze, puncture or drain it. Arrange medical assessment, particularly if it is growing, recurring, very painful or leaving scars. Seek urgent care if redness spreads, the area becomes hot and swollen, fever develops or you feel unwell."

15. Glossary and references

15.1 Glossary

Term	Definition
Acne vulgaris	Common inflammatory condition involving hair follicles and oil glands.
Comedone	Clogged follicle; open comedones are blackheads and closed comedones are whiteheads.
Papule	Small raised inflammatory lesion without obvious pus.
Pustule	Inflamed lesion with visible pus.
Nodule	Deeper inflammatory lump, often painful and more likely to scar.
Cyst	A sac-like lesion; a cyst-like skin lump needs proper diagnosis.
PIH	Post-inflammatory hyperpigmentation: dark marks after inflammation or injury.
Non-comedogenic	Formulated to be less likely to clog pores; not a guarantee for every user.
Retinoid	Vitamin A-related topical medicine used for acne and certain texture/pigmentation concerns.
Broad-spectrum sunscreen	Sunscreen that protects against UVA and UVB radiation.
Photoaging	Visible skin changes caused in part by cumulative UV exposure.
RAG	Retrieval-augmented generation: a system that retrieves source passages to ground generated answers.

15.2 References and further reading

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15.3 Evidence and scope note

This guide synthesizes public-facing guidance and clinical recommendations for education. Recommendations can change, product availability differs by country, and the best treatment depends on an individual's diagnosis, age, pregnancy status, medical history, skin tone, allergies, other medicines and preferences. Re-check source guidance before deploying this document in a public-facing clinical chatbot.

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